MIL-S-13065C
11 JULY 1960
SUPERSEDING
MIL-C-13065B
31 MAY 1957

*MILITARY SPECIFICATION*

# SHELL FOR CARTRIDGE, 4.2-INCH MORTAR, M2, M2 ALTERNATE, M2A1 AND M2A1 ALTERNATE; HD FILLING ASSEMBLY

*This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.*

## 1. SCOPE

1.1 This specification covers the mustard (HD) filling assembly (without fuzing components) of the M2, M2 alternate, M2A1, and M2A1 alternate 4.2-inch mortar shell for cartridge.

## 2. APPLICABLE DOCUMENTS

2.1 **Government documents.** The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

### SPECIFICATIONS

MILITARY

JAN-P-253 —Paper, Liquid-Vesicant-Detector, M6.

JAN-C-314 —Crayon, Vesicant (H or L) Detector, C14.

MIL-M-12051 —Mustard (HD) Vacuum Distillation Process.

### STANDARDS

MILITARY

MIL-STD-105 —Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 —Marking for Shipment and Storage.

### DRAWINGS

ORDNANCE CORPS

75-14-630— Shell for Cartridge, 4.2-Inch, Chemical, M2, M2 Alternate, M2A1, and M2A1 Alternate — Loading Assembly and Marking Diagram.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1315

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:01Z
Check the source to verify that this is the current version before use.

**2.2 Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

INTERSTATE COMMERCE COMMISSION

49 CFR 71-90—Interstate Commerce Commission Rules and Regulations for the transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (Revised 1956) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71-90 (Rev. 1956).")

## 3. REQUIREMENTS

**3.1 Materials and components.**

3.1.1 *Materials.* All materials cited on drawing 75-14-630 or on subsidiary drawings shall conform to the specifications thereon, or to the specific characteristics set forth on the drawings.

3.1.2 *Components.* All components of the shell for cartridge shall conform to the specifications and drawings listed on drawing 75-14-630 and subsidiary drawings.

**3.2 Filling and assembly.**

3.2.1 *Filling.* The shell for cartridge, dry and free from foreign matter, shall be filled with HD conforming to Specification MIL-M-12051 (see 6.2).

3.2.2 *Assembly.* Closing, coating, marking, and assembly of the shell for cartridge and the interior of the burster casing shall be thoroughly cleaned to remove any grease, oil, agent, or other contamination (see 6.4).

**3.4 Leakage.** The cleaned shell for cartridge shall not leak when tested as specified in 4.5.1.

**3.5 Filling height.** The distance from the surface of the HD in the filled and closed shell for cartridge to the bottom of the burster well adapter shall be within the limits of figure 1 when the cartridge is inspected in a vertical noseup position at the corresponding temperature range specified thereon and when tested as specified in 4.5.2 (see 6.3).

**3.6 Workmanship.** The shell for cartridge shall be free from contamination (foreign matter) and damage such as dents, cracks, or any deformation which would affect its use.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

**4.2 Objective evidence.** The supplier shall present objective evidence as required by the Government representative that all the requirements of sections 3 and 5 have been met (see 6.6).

**4.3 Alternative inspection (including testing) procedures.** The supplier may utilize any alternative inspection procedure which will assure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to its institution. In case of dispute, the procedures of this specification shall govern.

**4.4 Inspection provisions.**

4.4.1 *Regular production lot.* A regular production lot shall consist of the shell for cartridge filled by one manufacturer, at one plant,



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:01Z
Check the source to verify that this is the current version before use.

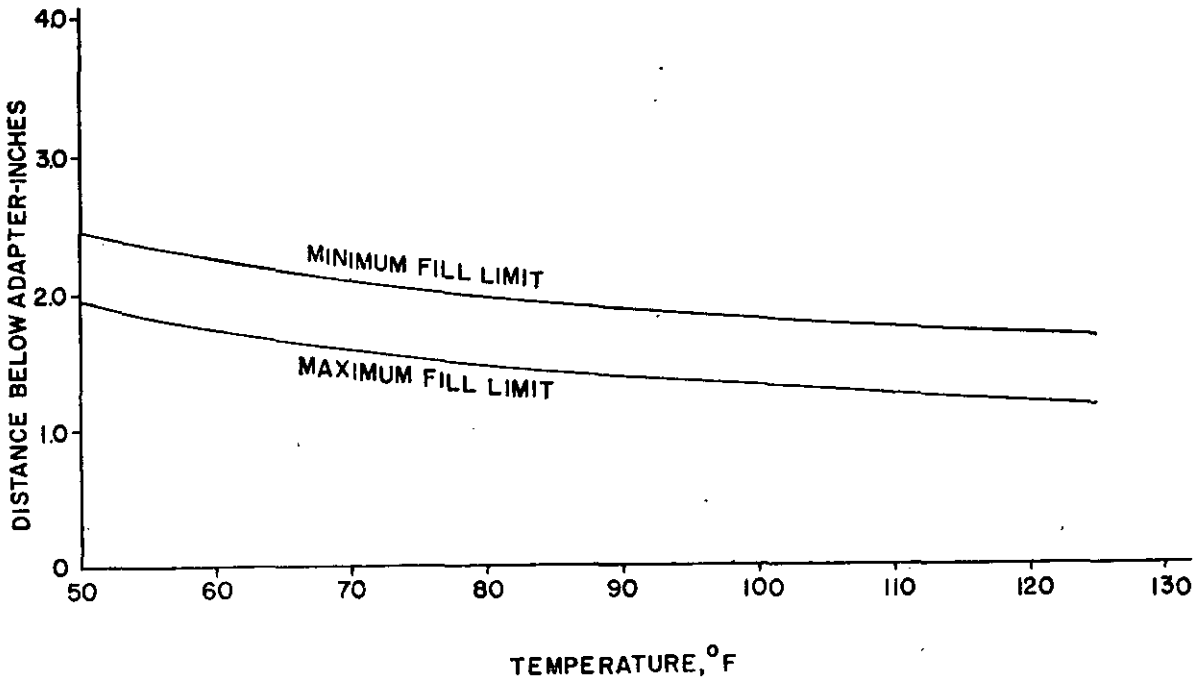



FIGURE 1. *Inspection criteria curves for closed cartridge, 4.2-inch mortar, M2, M2 alternate, M2A1 and M2A1 alternate.*

from the same materials, under essentially the same filling conditions and offered for delivery at one time. However, not more than one lot of HD filling shall be represented in any one lot of filled shell for cartridges.

4.4.2 *Sampling.* Sampling shall be conducted in accordance with Standard MIL–STD–105.

4.4.3 *Examination.* Sample shell for cartridges shall be examined in accordance with the classification of defects and with Standard MIL–STD–105.

4.4.4 *Classification of defects.*

4.4.4.1 Shell for cartridge, 4.2-inch chemical, M2, M2 alternate, M2A1, and M2A1 alternate — loading assembly and marking diagram (dwg. 75–14–630).

Each shell for cartridge in the lot shall be examined for the defects.

| *Categories* | *Defects* | *Acceptance standards* |
|---|---|---|
| *Critical*: | | |
| 1 | Leakage | 4.5.1 |
| 2 | Filling excessive | 4.5.2 |
| 3 | Agent contamination. | |
| *Major*: | *AQL 1.0 percent defective* | |
| 101 | Burster casing incorrectly assembled. | |
| 102 | Threads damaged or contaminated. | |
| 103 | Bourrelet diameter after coating incorrect. | |
| 104 | Marking incorrect. | |
| 105 | Coating missing, inadequate, or incorrect. | |
| *Minor*: | *AQL 2.5 percent defective* | |
| 201 | Marking missing or illegible. | |
| 202 | Filling insufficient | 4.5.2 |
| 203 | Damage (see 3.6). | |
| *Major A*: | *None defined* | |
| *Incidental*: | *AQL 10.0 percent defective* | |
| 401 | Contamination (other than agent). | |



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:01Z
Check the source to verify that this is the current version before use.

4.5 **Tests.**

4.5.1 *Leakage test.* After the final cleaning operation, a disk of coated paper shall be inserted into the fuze well so that the coated side of the paper is in contact with the burster adapter. The disk shall be cut either from detector paper conforming to Specification JAN–P–253 or from any other absorbent paper. The absorbent paper shall be coated with crayon conforming to Specification JAN–C–314. The shell for cartridge shall be maintained on its side or in a nosedown position at a minimum temperature of 70° F. for a minimum period of 16 hours and the disk examined for discoloration which shall be considered evidence of leakage.

4.5.2 *Filling height.* The height of fill shall be determined in accordance with Inspection Standard No. R–13–6 (see 6.1).

## 5. PREPARATION FOR DELIVERY

5.1 **Level A packing.** Unless otherwise specified, the filled and closed shell for cartridges shall be packed for shipment or storage in the same manner as the empty shell for cartridges were received and the pack shall conform to the ICC regulations. Shell for cartridges shall be segregated by lot for shipping purposes.

5.2 **Marking.** In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with Standard MIL–STD–129 and figure 2.

## 6. NOTES

6.1 **Ordering data.** Procurement documents should specify:

(a) The title, number, and date of this specification.

(b) The shipment group size and marking (when required).

6.2 **Precautions.** Because of the hazardous nature of filling munitions with chemical agents, filling installations are warned that all possible precautions be taken to protect personnel from injury while filling shell for cartridges in accordance with this specification.

6.3 **Filling aid.** Shell for cartridges filled with HD to a level below the top of the nose before insertion of the burster casing for the corresponding temperature shown in figure 3, have been found to comply with 3.5 provided the shell for cartridges meet the dimensional requirements of the applicable drawings. This height of fill is equivalent to a 14 ± 2 percent void. The HD filling in the shell for cartridge will weigh 5.74 ± 0.18 pounds. This information is furnished as an aid to procurement and not as a criterion for filling.

6.4 **Decontamination.** A shell for cartridge that becomes contaminated with HD during the filling and closing operations may be decontaminated with caustic solution followed by a hot water or steam rinse. Other approved decontamination solutions may be used.

6.5 **Safety classification.** This item is group A chemical ammunition and carries the following Ordnance Corps safety classification: storage compatibility group A, explosive hazard class 11.

6.6 **Objective evidence.** Provisions for objective evidence and inspection records and maintenance of inspection records will be specified by the procuring activity.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:01Z
Check the source to verify that this is the current version before use.

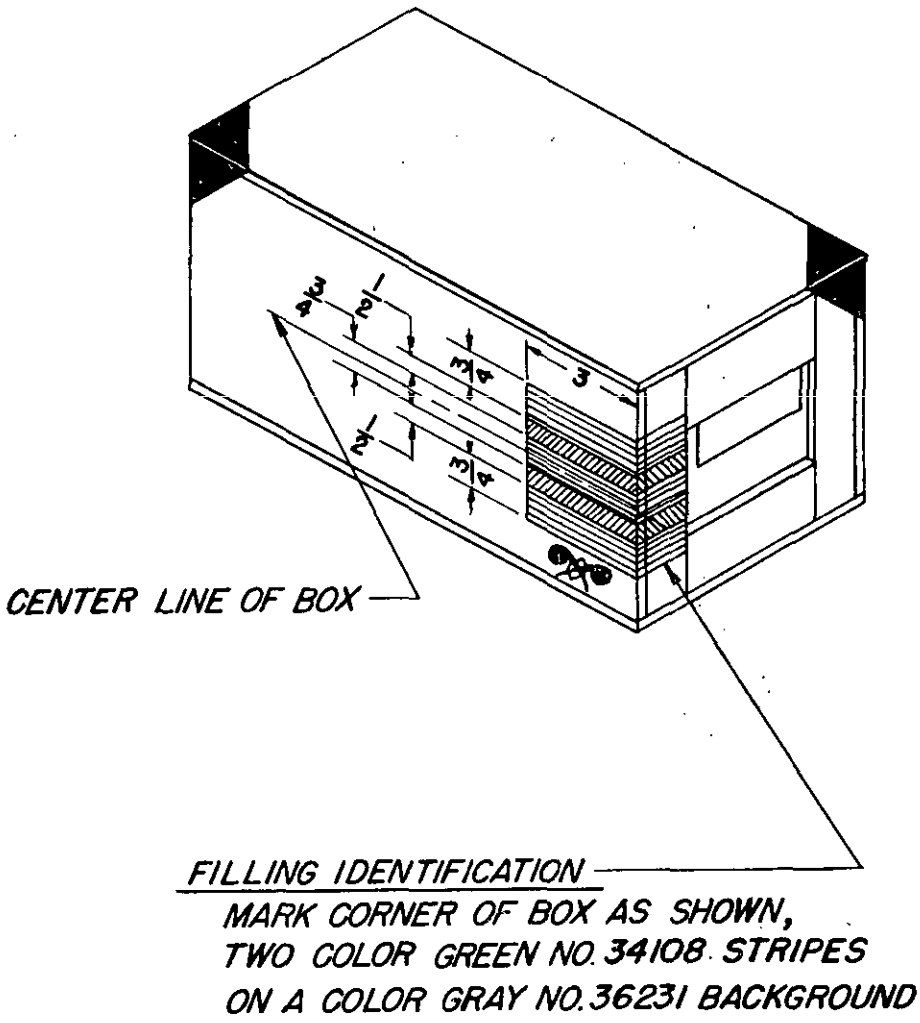



FIGURE 2. *Filling identification.*



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:01Z
Check the source to verify that this is the current version before use.

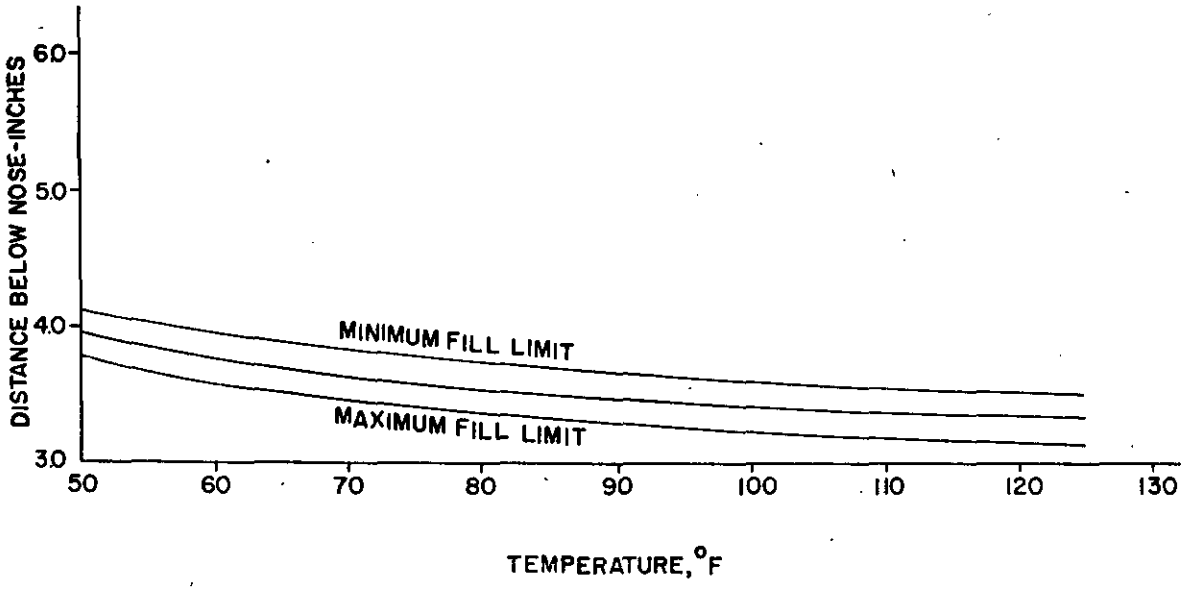



FIGURE 3. *Filling aid curves for open cartridges, 4.2-inch mortar, M2, M2 alternate, M2A1 and M2A1 alternate.*

**Notice.** When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

**Custodians:**
**Army—Chemical Corps**
**Navy—Marine Corps**

**Preparing activity:**
**Army—Chemical Corps**



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:01Z
Check the source to verify that this is the current version before use.